

LESSON 5.4

POINTS ON THE COORDINATE PLANE – PART 2



LESSON INTRODUCTION

MA.6.GR.1.1 Extend previous understanding of the coordinate plane to plot rational number ordered pairs in all four quadrants and on both axes. Identify the x- or y-axis as the line of reflection when two ordered pairs have an opposite x- and y-coordinate.

LEARNING TARGETS

- I can plot rational numbers using a scale of 1.

BENCHMARK COHERENCE

BUILDING ON

MA.5.GR.4.1 Identify the origin and axes in the coordinate system. Plot and label ordered pairs in the first quadrant of the coordinate plane.

WORKING TOWARD

N/A

ESSENTIAL QUESTIONS

- How can you plot decimal coordinates on a graph with a whole-number scale?
- How can you plot fractional coordinates on a graph with a whole-number scale?

LESSON NARRATIVE

Students continue plotting coordinates with rational numbers in this second part of a two-part lesson series. Students apply what they have learned in engaging, game-like contexts in addition to real-world examples. Students will need to analyze the data to determine the range of values and the best location of the axes and plot rational numbers with a scale of one.

COMPONENT SUMMARY

5.4.1 WARM-UP (~5 MIN.)

Compare different graphs of the same data

5.4.2 TRY IT (10 MIN.)

Graph rational numbers

5.4.3 COLLABORATION (15-20 MIN.)

Pinpoint coordinates playing a graphing game

5.4.4 YOUR TURN! (10-15 MIN.)

Graph rational numbers and compare coordinates

5.4.5 WRAP UP (~5 MIN.)

Reflect on understanding of past four lessons and discuss next steps

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Coordinate

MATERIALS

- (Optional) Computer

ADDITIONAL PREPARATION

- The gameboard in 5.4.2 Try It is pre-made and available to print.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle with estimating the location of rational ordered pairs. When graphing negative fractions and decimals on a whole-number scale, students may struggle with which number to graph closer to.
- Improper fractions may be incorrectly graphed.

THINGS TO CONSIDER

- Students are still grasping negative numbers and may need vertical and horizontal number lines to aid their thinking.
- Students may struggle with converting improper fractions to mixed numbers, which may cause a roadblock to graphing. Consider having students convert all mixed numbers before beginning to graph.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

Students participate in a Notice and Wonder routine during this lesson's Warm-Up. This positions students to consider how the scales of graphs impact the way data is presented visually and encourages students to draw conclusions about a data set from a graph.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

5.4.4 Your Turn! positions students to consider how information in the real world can be represented using graphs and coordinate planes. Students graph data relating to temperature in two cities and then NBA player statistics. Encourage students to draw conclusions about the significance of graphing data in the real world.

DIFFERENTIATE INSTRUCTION

Throughout this lesson, students engage with the content in a variety of ways. As students graph data points on the coordinate plane, they visually represent the data. The 5.4.2 Try It and 5.4.3 Collaboration activities encourage partner discussion for students to process the learning verbally, providing auditory entry points for learners as well.

5.4.1 WARM-UP: TEAM WINS

Component Overview: In this Warm-Up, students will discuss what they notice and wonder about two graphs representing the same data set.

TEACHER MOVES

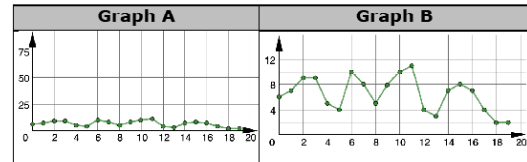
Notice and Wonder

Engage the class in a Notice and Wonder routine. Position students to discuss what they see on each graph, ensuring they understand both graphs represent the same data set and information. Students should begin to draw conclusions about the impact of scale on the graphs and consider which graph displays the data in a more understandable way.



5.4.1 Warm-Up: Team Wins

Each graph represents the number of wins for the Arkansas Razorbacks football team for the 2000-2019 seasons, where the x-axis represents the number of years since 2000.



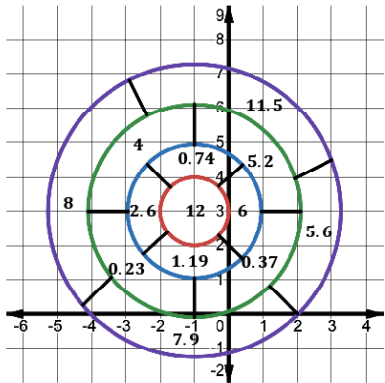
- What do you wonder about the two graphs?**
Answers will vary. I wonder why the changes between the points in Graph A don't seem as big as they are in Graph B.
- What do you notice about the two graphs?**
The scale for the y-axis is different between the two graphs.
- In what years did the team win at least 10 games?**
Years 6, 10, and 11
- Explain how the scale of the y-axis impacts the graphs.**
Answers will vary. The scale of the y-axis can make the data more or less clear. If the scale used is too big, like in Graph A, it is hard to get exact data and to see the trends of the data.



5.4.2 Try It: Game Score

A gameboard is shown on the coordinate grid, along with the number of points earned for landing in each area. Each coordinate grid axis is scaled at 1.

1. Several points are given in the table. Locate each point on the coordinate grid. Complete the table by writing the score value for landing at each point. Calculate the game score by finding the sum of all the points. Verify your answer with your teacher.



Point	Score Value
$(-3, 4.9)$	4
$(\frac{4}{5}, 1\frac{3}{4})$	0.37
$(0, 2.2)$	6
$(-3.8, 4.9)$	8
$(1.5, 4.4)$	5.2
$(-1\frac{2}{3}, 2)$	1.19
$(\frac{5}{4}, \frac{11}{2})$	11.5

Game Score: 36.26

5.4.2 TRY IT: GAME SCORE

Component Overview: In this Try It, students will plot rational-number coordinates and determine the value of that part of the game board. Students need to plot fractional and decimal coordinates on the given whole-number scale.

DIFFERENTIATE INSTRUCTION

Consider providing additional visual support for students, as the graph may be difficult to read given the number of point values and the scales on the x- and y-axes. It may be helpful to project the game board for the class so students can more clearly see where each point value is in relation to the scales, and/or to color-code each point value with the associated region of the gameboard.

ENRICHMENT

If students are struggling to plot rational numbers on the gameboard, consider further scaffolding through questioning like:

- How can you plot decimal coordinates on a graph that uses a whole-number scale? How can you plot fractions on a graph that uses a whole-number scale?
- Do the coordinates need to be plotted accurately in this activity? Why?

5.4.3 COLLABORATION: HIT OR MISS

Component Overview: In this Collaboration, students will play a Battleship-style game. Each student chooses a shape to plot on their coordinate plane. Each partner guesses a point on the plane to try and determine the shape their partner chose.

TEACHER MOVES

Encourage partners to engage in mathematical discussion regarding their strategies.

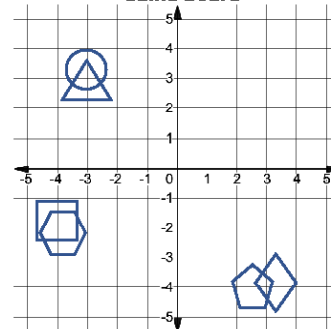
- Who won? Was their strategy the same as yours?
- Did you have any issues plotting points or coming up with ordered pairs? Why?
- How did your knowledge of graphing points help you in this game?
- How did the scale of the axes impact the guessing game?



5.4.3 Collaboration: Hit or Miss

There are six shapes on the coordinate grid shown. Each partner will select one shape as their game piece. The object of the game is to guess which shape your partner selected before they guess yours. A hit means the point you guessed is inside their shape. A miss means that it is not. Keep your game board hidden from your partner during the game.

Game Board



1. The shape of the region I am picking is a _____.

Answers will vary.

2. Record the coordinates of each guess in the table below. Then, plot the point on the coordinate grid game board.

Answers will vary. See example below

Partner 1 Guess	Was it a hit or miss?	Partner 2 Guess	Was it a hit or miss?
(3,5)	Miss	(-3,2)	Miss

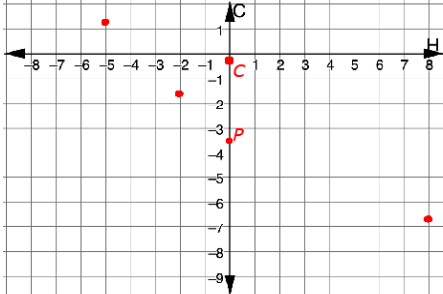


5.4.4 Your Turn!

1. The temperature in Princeton, New Jersey was recorded at various times during the day. The times relative to midnight, H , and temperatures in degrees Celsius ($^{\circ}\text{C}$) are shown in the table.

Time (hours before or after midnight, H)	Temperature ($^{\circ}\text{C}$)
-5	$\frac{6}{5}$
-2	$-1\frac{3}{5}$
0	$-\frac{7}{2}$
8	$-6\frac{7}{10}$

- a. Plot the points that represent the data.



- b. In the town of New Haven, Connecticut, the temperature at midnight was $-\frac{2}{10}^{\circ}\text{C}$. Plot and label this point on the same grid. Label the point C.
- c. Label the point that represents the temperature in Princeton at midnight P. Which town was warmer at midnight, Princeton or New Haven?
New Haven is warmer.

5.4.4 YOUR TURN!

Component Overview: In this Your Turn!, students will plot data on a graph based on real-world contexts. Students plot positive and negative rational numbers on the coordinate plane and use the graphs to draw conclusions about the data set.

ENRICHMENT

Students may graph negative fractions and mixed numbers. Consider posing questions to help students reason about where to plot the points, such as:

- How did you know where to plot the fractions whose denominators were more or less than halfway?
- If the time is in hours before or after midnight, what time is zero?
- Why is 2 the highest number on the y-axis?

ENRICHMENT

If students struggle to answer questions 1c and 1d, consider asking the following to provide scaffolding:

- Which coordinates should be compared to determine the warmer town?
- Is a negative mixed number greater or less than a negative fraction?
- As temperatures go further negative, are they getting hotter or colder?

5.4.4 YOUR TURN! (CONTINUED)

DIFFERENTIATE INSTRUCTION

Students can label each column “x” and “y” or write out ordered pairs to help remember which axis each number is on.

TEACHER MOVES

MA.K12.MTR.7.1: Mathematics in Real-World Context

Students are graphing specific NBA players’ average points per game and how much they deviate from the average of the rest of the league. Consider asking:

- What does it mean when a coordinate is close to the x-axis on the graph?
- What does it mean when a player’s coordinates are higher on the graph?
- What does it mean when a player’s coordinates are lower on the graph?

QUESTIONING

Have students who finish early answer the following:

- What jobs do you think use this type of data?
- How would this graph be helpful to them?
- Are there jobs that you are interested in?

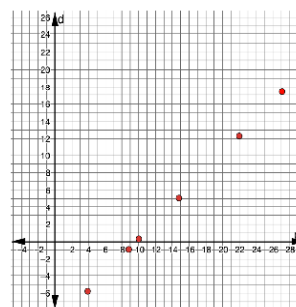
- d. Explain how to determine which town you selected in Part C is warmer at midnight.

Using the y-coordinate of both points, $-\frac{2}{10}$ is further up or greater than $-\frac{7}{2}$ meaning that it is warmer the other number.

2. The average NBA player scores 9.7 points per game. The table below lists six players, along with their average points per game and their difference from the average.

Player	Average Points Per Game (p)	Difference from Average (d)
LeBron James	27.1	17.4
Dwyane Wade	22.0	12.3
Mike Conley	14.8	5.1
Frank Brickowski	10.0	0.3
Robin Lopez	8.8	-0.9
Greg Foster	3.9	-5.8

Graph the points from the table on the coordinate grid.





5.4.5 Wrap-Up: Reflection

Think about what you have learned in the past four lessons about plotting points. As you read each statement, consider which best describes your understanding right now.

1. I need help plotting points. I'm not sure where to begin.
 2. I can plot points with integers on a coordinate grid with a scale of 1, but I still need help with rational numbers and different scales.
 3. I can plot points with decimals on a coordinate grid with a tenths scale, but I still need help with points that are fractions.
 4. I can plot points with rational numbers that are decimals or fractions using any scale.
1. Explain which number from the list best describes you.
Answers will vary.
 2. Write a plan for what you can do to either move to number 4 or to help a classmate move to number 4.
Answers will vary. I will practice labeling and plotting ordered pairs on a graph to increase my understanding and accuracy.

5.4.5 WRAP-UP: REFLECTION

Component Overview: In this Wrap-Up, students will reflect on their understanding on a scale from one to four to determine what they understand from the past four lessons and how they can improve if their understanding is less than a four. Consider using this reflection to help target who to pull for a small group.

DIFFERENTIATE INSTRUCTION

Students may need to put either a checkmark or an "X" by each statement they agree or disagree with to help frame their thinking. When answering question 2, consider giving students a sticky note to write their answer on and place on their desk to help remind themselves how to achieve their goal.